

Notes on *Credit Supply and House Prices: Evidence from Mortgage Market Segmentation*

(Adelino, Schoar, Severino; Journal of Financial Economics, 2025)

Study Notes (prepared as a coherent review sheet)

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1 What this paper is about

Core question. How much of an *exogenous* change in mortgage credit availability and cost is capitalized into house prices, holding constant local economic conditions and house-price expectations?

Core contribution (identification). The paper builds a **local difference-in-differences (DiD)** estimator that leverages:

- annual changes in the **conforming loan limit (CLL)**, and
- the industry-standard **80% loan-to-value (LTV) threshold**,

to generate sharp micro-level variation in whether a given transaction can be financed with an 80% LTV conforming mortgage. The main price result is that houses that become newly “easy to finance” (eligible for an 80% conforming loan) increase in value by about **\$1.17 per square foot**, with an implied **local elasticity of house prices to interest rates below 6**.

Why this matters conceptually. Macro correlations between credit conditions and house prices face a causality problem: credit expansions may be *caused by* rising prices and optimism (collateral feedback and expectations), not only the other way around. This paper aims to isolate a *credit-supply component* that shifts the user cost of borrowing for a narrow slice of the housing market, in a way that is plausibly orthogonal to local conditions.

2 Institutional background: conforming vs. jumbo and the 80% LTV rule

Conforming loan limit (CLL). The CLL is the maximum mortgage size eligible for purchase/securitization by the GSEs (Fannie Mae, Freddie Mac). Conforming loans tend to be cheaper and easier to obtain than “jumbo” loans (above the limit), due to implicit/explicit government support and different screening intensities.

Why the 80% threshold matters. In practice, mortgages with $LTV \leq 0.8$ avoid private mortgage insurance and receive better terms. Hence, the combination of:

$$(i) \text{ loan size} \leq \text{CLL} \quad \text{and} \quad (ii) \text{ LTV} \leq 0.8$$

creates a salient segmentation in the mortgage market.

House-price threshold used by the paper. Define the transaction price threshold as:

$$\text{Threshold}_t \equiv \frac{\text{CLL}_t}{0.8} = 1.25 \cdot \text{CLL}_t.$$

If a house transacts at $P_{it} \leq 1.25\text{CLL}_t$, then it can be financed with an 80% LTV *conforming* mortgage. If $P_{it} > 1.25\text{CLL}_t$, an 80% first-lien mortgage would exceed CLL_t , forcing the borrower to either:

- take a jumbo loan (higher interest rate and tougher screening), or
- keep the first lien at (or below) CLL_t and increase the down payment / use costly alternative funding.

3 Benchmark theory: the user-cost model

Purpose of this section. The paper uses the standard user-cost logic to frame why interest-rate elasticities can be large *in theory* but hard to identify *in data* without clean exogenous variation.

3.1 User-cost indifference condition (Equation (1))

Let R_t be rent in period t , P_t the house price, r the mortgage interest rate, ϕ the tax rate relevant for deductibility, τ property taxes and maintenance (as a share), and g the growth rate of rents/maintenance (and, in steady-state versions, the relevant growth/appreciation term). Under a simplified steady-state with constant rates and $\rho = r$, the indifference condition is:

$$\frac{R_t}{P_t} = (1 - \phi)r - g + \tau. \quad (1)$$

Interpretation. The price-to-rent ratio is the inverse of the user cost. Lower mortgage rates decrease the user cost and, in the simple model, raise P_t strongly.

Elasticity intuition. The paper illustrates that small changes in r can imply large changes in P/R under the simplest assumptions, but more realistic assumptions (e.g., discount rates not equal to market rates; mean-reverting rates; prepayment) can generate substantially smaller elasticities. This motivates directly estimating capitalization *locally* using micro shocks.

4 A simple financing-cost model around the threshold (mechanism intuition)

Key point. The CLL creates a *bundled treatment*: above 1.25CLL borrowers face a higher *weighted average* cost of funding, but the intensity depends on their outside options (access to jumbo market; cost of additional down payment funding).

4.1 Weighted average annual cost comparison

Write the transaction price as $P = m \cdot \text{CLL}$ (so $m = 1.25$ is the threshold). Consider financing an 80% LTV position.

Option A: conforming loan up to the limit + extra down payment. Let r_c be the conforming rate and r_{DP} the marginal cost of additional down-payment funds (from savings, second liens, etc.). If the borrower takes the maximum conforming first lien (CLL) and funds the remaining amount needed to reach 80% of value via down payment, the annual dollar cost is:

$$r_c \cdot \text{CLL} + r_{DP} \cdot (0.8m \text{CLL} - \text{CLL}).$$

Option B: jumbo first lien at 80% LTV. Let r_j be the jumbo rate. The annual dollar cost is:

$$r_j \cdot 0.8m \text{CLL}.$$

4.2 Jumbo vs. conforming+DP choice condition

Define the jumbo-conforming spread $r_{j-c} = r_j - r_c$ and the down-payment spread $r_{DP-c} = r_{DP} - r_c$. Comparing the two annual costs implies the borrower prefers a jumbo loan if:

$$r_{j-c} \leq r_{DP-c} \cdot \left(1 - \frac{1}{0.8m}\right).$$

Interpretation.

- If r_{DP} is very high, even a modest jumbo spread makes jumbo relatively attractive.
- If a borrower is excluded from jumbo credit, then their effective r_{DP} (or constraint) makes the wedge above the threshold grow steeply with m .

This heterogeneity is central for interpreting the treatment as a local credit-cost shock rather than a sharp discontinuity.

5 Data and measurement

5.1 Transactions data

The core dataset contains deeds/assessors records (Dataquick) for 1998–2008, covering 74 counties in 10 MSAs (Boston, Chicago, DC, Denver, Las Vegas, Los Angeles, Miami, New York, San Diego, San Francisco). The paper focuses primarily on single-family transactions (the large majority of observations).

5.2 Valuation measures

The paper uses three primary valuation outcomes in the main regressions:

- raw *value per square foot* = $\frac{\text{transaction value}}{\text{sq ft}}$,
- *log transaction value residual* from hedonic regressions,
- *value per square foot residual* from hedonic regressions.

6 Hedonic regressions for quality adjustment (Equation (2))

To separate price differences due to financing from those due to observable quality, the paper estimates hedonic regressions by MSA and by year:

$$LHS_i = \gamma_0 + \Gamma X_i + \text{month}_i + \text{zip}_i + \varepsilon_i, \quad (2)$$

where LHS_i is either value-per-sq-ft or log transaction value, and X_i is a rich vector of house characteristics. Residuals from these regressions form two alternative quality-adjusted valuation measures.

Why this step matters. If houses just above the threshold are systematically of lower quality than those just below, a naïve price comparison would confound financing effects with quality differences. Hedonic residuals provide a way to hold observed quality constant.

7 Identification and empirical design

7.1 Local DiD sample construction

For each year t (1998–2005), define the threshold $\text{Threshold}_t = 1.25\text{CLL}_t$. Construct a tight band around it:

$$[\text{Threshold}_t - 10,000, \text{Threshold}_t + 10,000].$$

The regression sample for cohort t includes:

- transactions in year t in that band, and
- transactions in year $t + 1$ in the *same dollar band*.

Because $\text{CLL}_{t+1} > \text{CLL}_t$ in these years, all transactions in the band become eligible for an 80% LTV conforming loan in year $t + 1$.

Treatment vs. control. Within cohort t :

- **Treatment group:** transactions with prices in $(\text{Threshold}_t, \text{Threshold}_t + 10,000]$ (“Above Threshold”) in year t .
- **Control group:** transactions with prices in $[\text{Threshold}_t - 10,000, \text{Threshold}_t]$ (“Below Threshold”) in year t .

In year t , treatment transactions are harder to finance (cannot do 80% conforming), while in year $t + 1$ *both* groups become easy to finance (eligible). This produces a clean DiD comparison.

7.2 Year-by-year DiD regression (Equation (3))

For each cohort year t , estimate:

$$\text{Valuation}_i = \beta_1 \mathbf{1}\{\text{Above Threshold}\}_i + \beta_2 \mathbf{1}\{\text{Year CLL}\}_i + \beta_3 (\mathbf{1}\{\text{Above Threshold}\}_i \times \mathbf{1}\{\text{Year CLL}\}_i) + \gamma \text{ZIP}_i + \varepsilon_i. \quad (3)$$

Coefficient of interest: β_3 . β_3 measures the relative valuation discount for the above-threshold group *in the year the CLL binds* (the “pre” year), compared to the same relative difference in the subsequent year when the limit is raised. A negative β_3 means harder-to-finance transactions sell for lower prices.

Aggregation across cohorts. The paper estimates Eq. (3) for each $t \in \{1998, \dots, 2005\}$ and reports Fama–MacBeth averages of coefficients, with standard errors computed from the time-series dispersion of the cohort estimates.

7.3 Pooled specification with event-cohort fixed effects (Equation (4))

As an alternative to Fama–MacBeth, the paper pools all cohorts and includes event-cohort fixed effects interacted with controls:

$$\begin{aligned} \text{Valuation}_{i,c} = & \beta_1 (\mathbf{1}\{\text{Above Threshold}\}_i \times \gamma_c) + \beta_2 (\mathbf{1}\{\text{Year CLL}\}_i \times \gamma_c) + \beta_3 (\mathbf{1}\{\text{Above Threshold}\}_i \times \mathbf{1}\{\text{Year CLL}\}_i) \\ & + \gamma (\text{ZIP}_i \times \gamma_c) + \nu_{i,c}, \end{aligned} \quad (4)$$

where γ_c is an event-cohort indicator (cohort c corresponds to the year in which the CLL is in effect for that cohort). Standard errors are clustered at the event-cohort level. The coefficient comparable to the Fama–MacBeth interaction estimate is again β_3 .

Why not a simple RDD within a year? Two reasons:

- The financing wedge is not a sharp discontinuity at exactly 1.25CLL: borrowers can “stretch” by adjusting down payments.
- Sorting is endogenous: households can choose to buy slightly below the cutoff to remain conforming, so being just below/above is not exogenous within a single year.

The DiD uses the exogenous *change in the limit* to generate identification.

8 Reading the evidence: figures and tables (interpretations)

In what follows, I interpret each table and each figure in the order they appear, and I connect them to the logic of the model and the DiD design.

8.1 Table 1: summary statistics (transactions and regression sample)

Table 1: Summary statistics.

	All transactions ($N = 3,956,516$)			Regression sample ($N = 261,404$)		
	Mean	SD	Median	Mean	SD	Median
<i>Panel A. House characteristics</i>						
Transaction value (\$1000)	308.79	124.04	287.00	371.39	54.92	380.00
Loan-to-value	0.81	0.15	0.80	0.76	0.13	0.80
House size (sq ft)	1733	671	1590	1932	700	1812
Lot size (sq ft)	9946	14,873	6671	11,496	17,381	7200
Number of rooms	6.84	1.60	7.00	7.22	1.62	7.00
Number of bedrooms	3.20	0.78	3.00	3.33	0.79	3.00
Number of bathrooms	1.93	1.03	2.00	2.11	1.08	2.00
House age (years)	35.41	27.59	34.00	34.81	27.32	34.00
<i>Panel B. House valuation</i>						
Value per sq ft (US\$/sq ft)	193.93	91.73	172.38	220.01	93.38	200.66
Value per sq ft residual (US\$/sq ft)	0.00	42.35	−0.96	5.26	44.30	3.40
Logarithm transaction value residual	0.00	0.17	0.01	0.05	0.14	0.04

Note. Panel A reports descriptive statistics for all transactions in the deeds data (1998–2008). Panel B reports valuation measures used in regressions. Residual measures come from hedonic regressions by year and MSA.

Table 1 documents two things:

1. The full dataset is large (millions of transactions), and median first-lien LTV is exactly 0.8, consistent with the institutional importance of the 80% threshold.
2. The regression sample is *more expensive and larger houses* (higher transaction values and sq ft), consistent with the idea that the CLL is designed to cover a substantial portion of the mortgage distribution.

8.2 Table 2: heterogeneity across MSAs and time

Table 2 provides two crucial pieces of background variation:

- Across MSAs, average prices and value per sq ft differ markedly (e.g., San Francisco high vs. Denver/Las Vegas lower), so the *same national CLL* bites in different parts of the local housing stock.

Table 2: Summary statistics by geography and year.

Panel A. Geographic distribution

MSA	No. obs.	Transaction value		Value per sq ft		Loan-to-value	
		Mean	SD	Mean	SD	Mean	SD
Boston	252,368	325.87	112.45	203.33	73.46	0.78	0.16
Chicago	376,974	262.39	108.14	174.37	68.63	0.81	0.15
DC	396,207	329.94	126.16	186.97	85.93	0.82	0.14
Denver	397,247	250.20	94.92	155.84	49.28	0.83	0.15
Las Vegas	345,218	262.24	102.87	136.62	45.38	0.82	0.14
Los Angeles	725,909	332.28	129.71	231.30	108.36	0.81	0.13
Miami	483,525	270.11	111.74	144.80	57.04	0.81	0.14
New York	487,020	340.99	121.13	221.25	92.55	0.78	0.17
San Diego	219,489	353.15	124.63	222.19	94.87	0.79	0.14
San Francisco	272,559	383.60	123.75	266.50	109.30	0.79	0.13
Total	3,956,516	308.79	124.04	193.93	91.73	0.81	0.15

Panel B. Distribution by year and thresholds

Year	No. obs.	Thresholds		Transaction value		Value per sq ft		Loan-to-value	
		House price	Conf. loan	Mean	SD	Mean	SD	Mean	SD
1998	133,347	283,938	227,150	240.06	102.20	134.13	50.61	0.81	0.15
1999	348,561	300,000	240,000	246.64	105.00	139.61	54.06	0.81	0.15
2000	351,611	315,875	252,700	257.94	109.35	149.94	61.73	0.81	0.16
2001	363,213	343,750	275,000	265.33	108.96	156.98	63.93	0.82	0.15
2002	394,586	375,875	300,700	283.98	114.48	171.33	71.99	0.81	0.15
2003	420,841	403,375	322,700	303.58	118.46	187.67	80.20	0.81	0.15
2004	522,003	417,125	333,700	332.05	121.31	212.97	90.62	0.79	0.14
2005	472,654	449,563	359,650	357.82	121.79	237.65	100.83	0.78	0.13
2006	373,881	521,250	417,000	366.66	121.92	247.47	105.59	0.79	0.13
2007	291,132	521,250	417,000	359.67	122.58	238.32	101.68	0.82	0.14
2008	284,687	521,250	417,000	325.37	119.90	207.25	91.71	0.84	0.15
Total	3,956,516			308.79	124.04	193.93	91.73	0.81	0.15

Note. Panel A reports means and standard deviations by MSA. Panel B reports year-by-year summary statistics and the thresholds relevant for the empirical analysis (house price threshold = $1.25 \times \text{CLL}$, and the CLL itself).

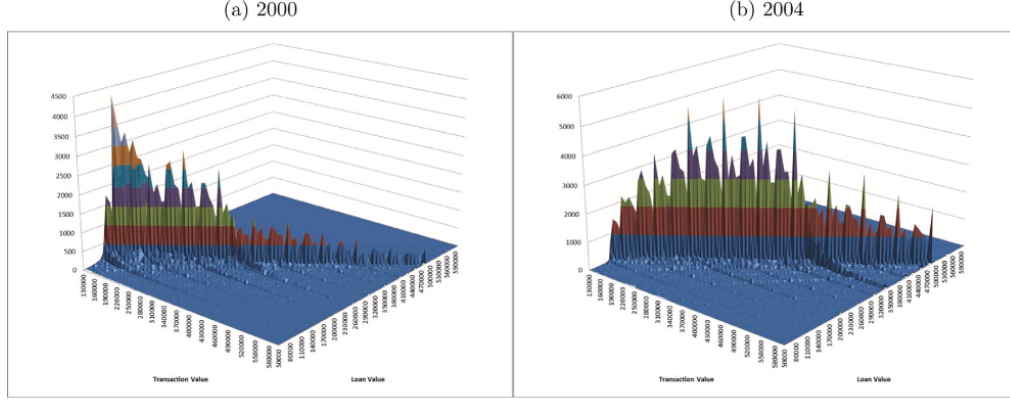


Figure 1: Transaction-loan value surface

- Over time, house prices rise sharply into 2006 and then fall; the CLL and threshold move over time as well. The average first-lien LTV stays around 0.8.

This heterogeneity helps rule out the interpretation that the result is driven by a single narrow type of borrower/house nationwide.

8.3 Figure 1: transaction-loan value surface (first-stage intuition)

Figure 1 shows the joint distribution of transaction values and loan values. My key takeaways:

- There is a thick mass on the diagonal corresponding to $LTV \approx 0.8$.
- There is a noticeable spike at the conforming loan limit: many borrowers choose a loan amount at the CLL and adjust down payment to keep the first lien conforming.

This motivates why a financing threshold can translate into discrete shifts in borrower behavior.

8.4 Figure 2 and Table 3: financing choices respond to CLL constraints

Figure 2 (composition). Figure 2 decomposes transactions near the threshold by LTV bins over time (year $t - 1$, year t when CLL binds, year $t + 1$ after the increase). The message is that in the binding year:

- below-threshold transactions cluster at 80% LTV,
- above-threshold transactions shift away from 80% LTV (either jumbo or conforming with larger down payment),
- in the subsequent year, the financing mix converges between the groups (consistent with both being eligible).

Table 3 (verification via regressions). Table 3 quantifies the financing wedge:

- *Loan-to-value*: the interaction term is negative (about -0.003 to -0.004), meaning that in the binding year, above-threshold transactions have lower first-lien LTV.
- *Loan amount*: the interaction term is also negative in log terms, consistent with borrowers staying at (or below) the conforming limit and using more equity/alternative financing.

This is the crucial first-stage: the policy threshold changes financing choices.

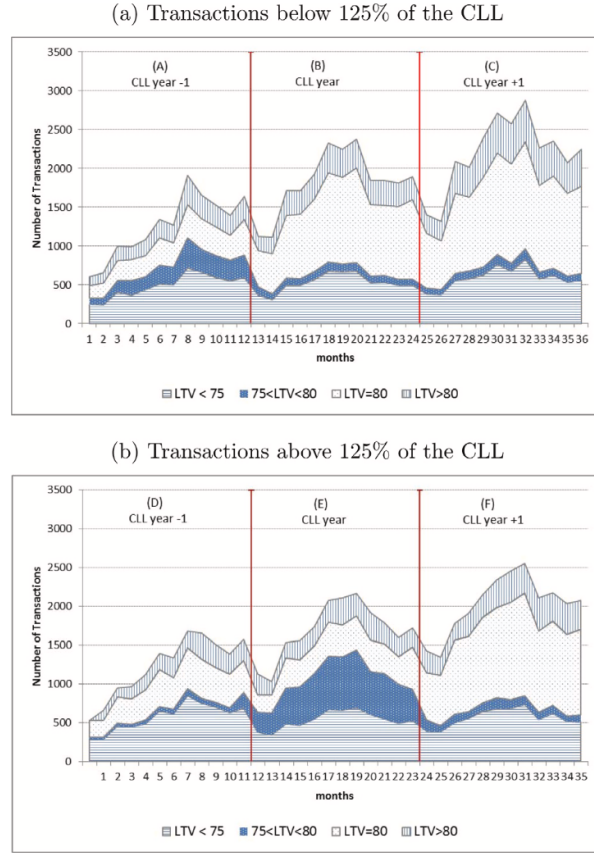


Figure 2: Borrower composition for the regression sample

Table 3: Verification of the impact of the CLL on financing choices.

	Loan-to-value	Log loan amount
<i>Panel A. Main specification (Fama–MacBeth)</i>		
Above Threshold	−0.004*** (0.001)	0.022*** (0.002)
Year CLL	−0.008*** (0.002)	−0.013*** (0.002)
Above Threshold × Year CLL	−0.004** (0.001)	−0.006** (0.003)
No. Obs.	241,598	241,598
<i>Panel B. Pooled specification</i>		
Above Threshold × Year CLL	−0.003** (0.001)	−0.005* (0.003)
No. Obs.	239,926	239,926

Note. The sample includes transactions within \$10,000 of each year’s threshold 1.25CLL and the same dollar band in the subsequent year. “Above Threshold” refers to transactions up to \$10,000 above 1.25CLL (ineligible for an 80% LTV conforming loan in the binding year). “Year CLL” is the year in which the limit is in effect (the “pre” year). Standard errors are in parentheses.

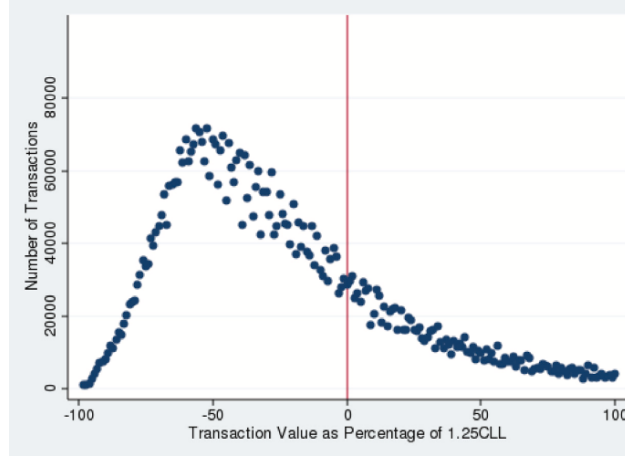


Figure 3: Frequency of transactions as a percentage of the CLL threshold

8.5 Figure 3 and Table 4: effects on transaction quantities

Figure 3. Figure 3 shows the distribution of transactions around the threshold (relative distance). There is no visible discontinuity right at the threshold, suggesting no sharp bunching or missing mass that would mechanically drive the price results.

Table 4: Impact of the CLL on the number of transactions.

	(1)	(2)	(3)	(4)
Year CLL	-0.003 (0.003)	-0.003 (0.003)	-0.016*** (0.004)	-0.017*** (0.003)
No. Obs.	261,404	261,404	261,404	259,699
ZIP code FE	N	N	Y	Y

Note. The dependent variable is an indicator for whether a transaction is above the threshold (within the \$10,000 band). Columns (1) and (3) report Fama–MacBeth coefficients from year-by-year regressions; columns (2) and (4) report pooled regressions with event-cohort fixed effects. Standard errors are clustered at the event-cohort level in pooled specifications. “Year CLL” is the year in which the conforming loan limit is in effect.

Table 4. Table 4 tests whether the *share* of transactions above the threshold changes between the binding year and the next year. The estimated effect is negative in the binding year (especially with ZIP fixed effects), indicating that constrained credit can reduce the number of transactions above the threshold. However, the paper emphasizes that the quantity effect is economically small overall relative to the price effect, which is consistent with low short-run housing supply elasticity: prices adjust more than quantities.

8.6 Table 5: the main price capitalization result

Table 5 is the centerpiece.

Table 5: Effect of the CLL on house valuation measures.

	Value per Sq. Ft.	Log of trans. value residual	Value per Sq. Ft. residual
<i>Panel A. Main specification (Fama–MacBeth)</i>			
Above Threshold	1.281*** (0.483)	0.013*** (0.001)	1.7540*** (0.3566)
Year CLL	−22.891*** (4.068)	0.039*** (0.004)	4.1475*** (0.6470)
Above Threshold × Year CLL	−1.173*** (0.269)	−0.002** (0.001)	−0.6689*** (0.2488)
No. Obs.	261,404	250,281	250,607
<i>Panel B. Treatment assignment robustness (predicted value assignment)</i>			
Above Threshold	−4.079*** (0.922)	0.003* (0.002)	−2.1821*** (0.5752)
Year CLL	−41.556*** (7.498)	0.006* (0.003)	−7.7823*** (2.2159)
Above Threshold × Year CLL	−0.948*** (0.228)	−0.003** (0.001)	−0.7923*** (0.2276)
No. Obs.	244,776	244,776	244,776
<i>Panel C. Pooled specification</i>			
Above Threshold × Year CLL	−1.123*** (0.279)	−0.002** (0.001)	−0.6160*** (0.2330)
No. Obs.	259,699	248,586	248,913

Note. Panel A reports Fama–MacBeth coefficients from year-by-year DiD regressions with ZIP fixed effects. Panel B assigns treatment using predicted values from hedonic regressions. Panel C pools cohorts and includes event-cohort fixed effects (see Eq. (4)); standard errors clustered at the event-cohort level.

Panel A (main specification). The interaction coefficient β_3 is negative across all valuation measures:

- about **-1.17** dollars per sq ft for raw value per sq ft,
- about **-0.002** in log transaction value residual,
- about **-0.67** dollars per sq ft for the quality-adjusted value-per-sq-ft residual.

I interpret this as: when the CLL binds, above-threshold houses are priced lower because they are harder/more expensive to finance; once the limit rises, the discount shrinks because the same price segment becomes eligible for conforming financing.

Quality adjustment. The drop from -1.17 to -0.67 when using residualized value per sq ft suggests that some of the raw difference is attributable to observable quality differences (houses just above the threshold may be of slightly worse quality), but a large and significant effect remains even after controlling for rich characteristics.

Panel B (treatment assignment robustness). Instead of using realized transaction price to define above/below threshold, the paper uses predicted values from a rolling hedonic model (see Eq. (5) below) to avoid concerns that treatment assignment mechanically correlates with the outcome. The interaction estimates remain similar, supporting the validity of the baseline approach.

Panel C (pooled specification). The pooled estimates are very close to the Fama–MacBeth interaction estimates, indicating that results are not an artifact of the two-step estimation.

8.7 Figure 4: heterogeneity by ZIP income (who is most constrained?)

Figure 4 plots value per sq ft against transaction value, separately by ZIP-code income quartiles.

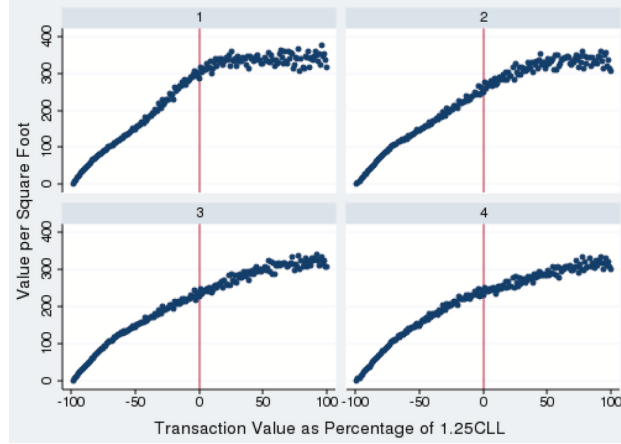


Figure 4: Value per square foot by house value and by ZIP code income

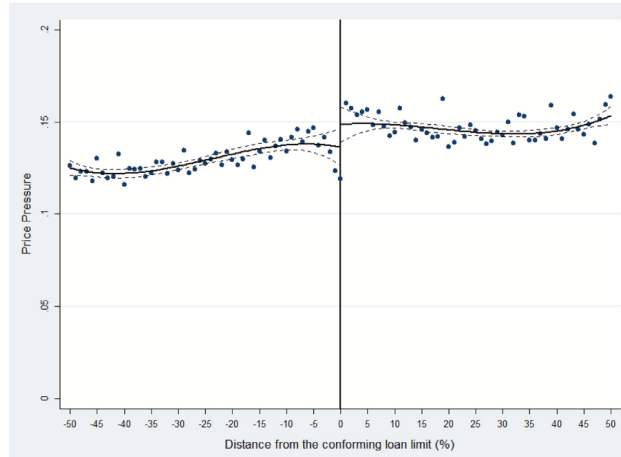


Figure 5: Share of unused mortgage applications

- The lowest-income quartile shows a visible “leveling off” around the threshold, consistent with the financing wedge binding more for constrained borrowers.
- Higher-income areas show a smoother pattern, consistent with better access to alternative financing or jumbo credit.

8.8 Figure 5: focal-point / bidding-pressure test

A concern is that the threshold is a focal point that attracts more bidders or changes listing behavior, which could confound interpretation. Figure 5 uses HMDA to measure the share of *unused/withdrawn* mortgage applications (a proxy for bidding pressure or failed offers). If the threshold were just a focal point for homes eligible for conforming loans, I would expect more unused applications below the limit (more bidders). Instead, the share is lower for conforming-amount loans, which supports the interpretation that the results reflect financing conditions, not just focal-point behavior.

Table 6: Placebo for coefficient of interest.

	All transactions			0.5 < LTV $\leq$ 0.8 transactions		
	Value per sq ft	Log trans. value residual	Value per sq ft residual	Value per sq ft	Log trans. value residual	Value per sq ft residual
True CLL	-1.173	-0.002	-0.669	-1.292	-0.002	-0.984
Placebo	0.056	0.001	0.228	-0.093	0.000	0.118
	(0.485)	(0.002)	(0.504)	(0.440)	(0.002)	(0.457)
<i>t</i> -statistic	2.534	1.275	1.780	2.725	1.048	2.414
CLL rank	1	3	2	1	3	1
CLL rank below only	1	2	1	1	1	1

Note. The paper shifts the true CLL by \$10,000 increments from CLL-100,000 to CLL+100,000 and recomputes the interaction coefficient each time. “Placebo” reports the average coefficient and standard deviation across the 20 placebo trials. Ranks compare the true CLL coefficient to placebo coefficients (overall and restricting to placebo limits below the true limit).

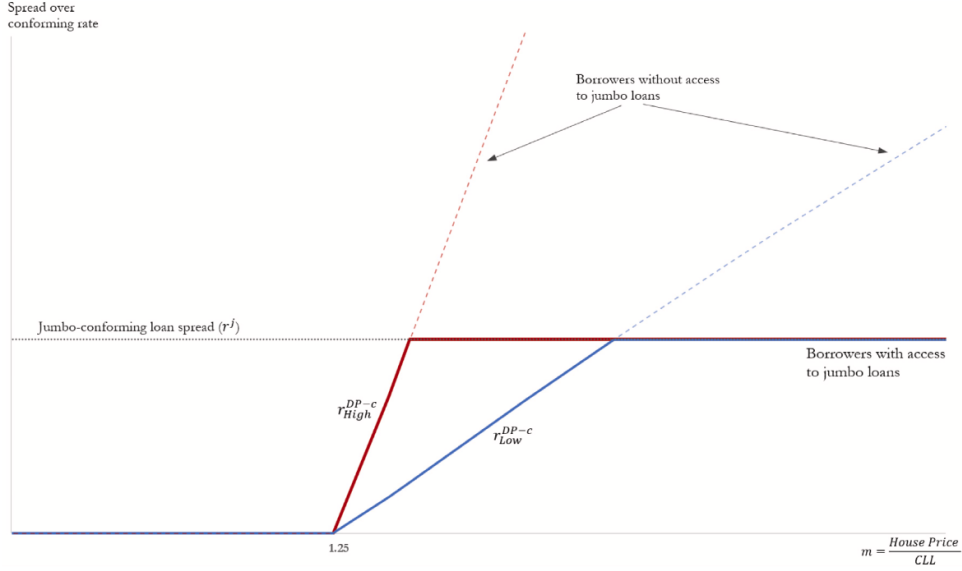


Figure 6: Illustration of weighted average spread over conforming loan rate

8.9 Table 6: placebo thresholds (falsification)

Table 6 shifts the CLL up and down in \$10,000 increments (placebos) and re-estimates the same DiD. The true CLL interaction coefficient is among the most negative across placebos (often rank 1), especially for the raw value-per-sq-ft outcome. This helps rule out the explanation that the estimate is driven by smooth differential price trends across the price distribution rather than by the CLL.

8.10 Figure 6: mapping the DiD into interest-rate semi-elasticities

Figure 6 illustrates how the cost-of-capital wedge increases above 1.25CLL, and why the treatment is bundled and heterogeneous.

Using the estimated price change (roughly 30–53 bps in price) and external estimates of the jumbo-conforming interest spread (roughly 10–24 bps), the paper computes local semi-elasticities of price with respect to interest rates that are below 6. This places their estimates at the lower end of the elasticity range implied by simple user-cost calibrations.

9 Treatment assignment robustness via predicted values (Equation (5))

To address concerns that realized prices mechanically affect treatment assignment, the paper estimates rolling hedonic models and assigns treatment using predicted values:

$$HouseValue_i = \theta_0 + \Theta X_i + month_i + ZIP_i + \varepsilon_i. \quad (5)$$

This alternative assignment produces similar results (Table 5, Panel B).

10 Economic magnitude: translating coefficients into price changes and elasticities

From dollars per sq ft to percentage price effects. Using the sample average of about \$220 per sq ft, the main interaction estimates imply:

$$\frac{1.17}{220} \approx 0.00532 \quad (53.2 \text{ bps}), \quad \frac{0.67}{220} \approx 0.00304 \quad (30.4 \text{ bps}).$$

Semi-elasticity to interest rates (local). Let $\Delta \ln P$ be the estimated percentage change and Δr the interest-rate wedge (jumbo-conforming spread). The semi-elasticity is:

$$\frac{\Delta \ln P}{\Delta r}.$$

Plugging $\Delta \ln P \in [0.0030, 0.0053]$ and $\Delta r \in [0.0010, 0.0024]$ yields a range roughly consistent with 1.3–5.3, and in any case below 6.

Interpretation. Even when financing becomes cheaper/easier, only part of the benefit accrues to buyers; a portion is capitalized into higher house prices, benefiting sellers/homeowners.

11 Final takeaways

- **Identification template:** use institutional segmentation + time variation in a policy threshold to build a DiD that is robust to endogenous sorting at a static cutoff.
- **Mechanism discipline:** show first-stage financing shifts (Figure 1–Figure 2, Table 3), show limited quantity response (Figure 3, Table 4), then interpret price effects (5).
- **Robustness toolkit:** predicted-value treatment assignment (Eq. (5)), placebo thresholds (Table 6), and focal-point tests (Figure 5).
- **Economic interpretation:** treat the wedge as a bundled treatment (rate + access + screening), and be transparent about mapping to interest-rate elasticities.

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Reading Note (Conclusion + Intuition):

Credit Supply and House Prices: Evidence from Mortgage Market Segmentation

Adelino, Schoar, and Severino (JFE 2025) Personal Study Notes January 27, 2026

Summary (What the paper does)

Question. How much does easier access to mortgage credit (and lower mortgage financing costs) get capitalized into house prices, holding fixed local economic conditions and local housing-market fundamentals?

Identification challenge. Aggregate movements in mortgage rates or credit conditions are endogenous to housing expectations, local demand, and collateral feedback loops. Therefore, identifying the causal effect of credit access/cost on house prices requires quasi-exogenous, *micro-level* variation.

Key idea: mortgage market segmentation around the conforming loan limit (CLL) at 80% LTV. Because conforming mortgages are subsidized and more easily available than jumbo mortgages, there is a sharp difference in the marginal cost of funding for transactions that can be financed with an 80% LTV conforming loan versus transactions that cannot. The relevant price threshold is:

$$P_t^* \equiv \frac{\text{CLL}_t}{0.8} = 1.25 \cdot \text{CLL}_t.$$

Transactions just above P_t^* face a higher weighted-average cost of capital: borrowers either (i) switch to a jumbo loan (higher rate), or (ii) keep a conforming loan at the CLL and finance the remainder with a larger down payment and/or other higher-cost sources.

Research design: difference-in-differences using annual changes in the CLL. A simple regression discontinuity around P_t^* is not appropriate because borrowers may strategically adjust loan size and down payments to remain at/below the CLL. Instead, the paper uses year-to-year increases in CLL_t (which shift P_t^*) to construct a “before/after” comparison for the *same narrow band of transaction prices*. Intuitively:

- In the “pre” year, houses just above P_t^* are harder to finance at 80% LTV with conforming loans.
- In the subsequent year (after the CLL rises), those same-price houses become eligible for an 80% LTV conforming loan.
- Comparing price changes for the above-threshold group to the below-threshold group differences out local market trends and time-invariant neighborhood characteristics.

Main Estimating Equation (the object that pins down the conclusion)

In each CLL event-year, the baseline specification is:

$$\text{Valuation}_i = \beta_1 \text{AboveThreshold}_i + \beta_2 \text{YearCLL}_i + \beta_3 (\text{AboveThreshold}_i \times \text{YearCLL}_i) + \gamma_{\text{ZIP}} + \varepsilon_i.$$

The coefficient of interest is β_3 : it measures the valuation gap between above- and below-threshold houses *in the year when the CLL binds*, relative to the subsequent year when both groups are eligible for 80% LTV conforming financing.

Conclusion (What the paper concludes)

- **Easier access to conforming credit raises house prices.** When a house becomes easier to finance with an 80% LTV conforming loan due to an increase in the CLL, its value increases by about **\$1.17 per square foot** relative to a closely comparable house just above the eligibility threshold. With an average value per square foot around **\$220** near the threshold, this corresponds to about a **0.53%** increase in prices.
- **Borrowers respond strongly to the CLL constraint, which is why a simple RDD is not valid.** Most borrowers target an 80% LTV, but many deviate (e.g., by increasing down payments) to keep the first mortgage at or below the conforming loan limit when the transaction price crosses $1.25 \times \text{CLL}$. This “stretching” behavior implies that treatment assignment is not sharp at the threshold.

- **The price effect reflects capitalization of financing advantages, not a pure focal-point bidding effect.** The paper argues that the estimated price differences are consistent with a cost-of-credit mechanism and performs placebo-style falsification exercises to support that interpretation.
- **The implied local elasticity of house prices to interest rates is relatively muted.** Using external estimates of the jumbo-conforming spread, the paper translates the estimated price response into a *local* elasticity range of roughly **1.3 to 5.3** (i.e., below 6), suggesting that interest-rate changes are capitalized into prices less than what some aggregate/time-series approaches imply.
- **Incidence: part of the conforming loan subsidy accrues to sellers.** Because easier access to subsidized conforming credit increases transaction prices, some of the benefit of the subsidy is passed through into higher house prices (higher seller-side surplus), not only lower borrowing costs for buyers.

Intuition (Why the result is economically natural)

Intuition 1: user-cost logic + segmentation creates a discrete change in the marginal bid

A standard user-cost framework implies that housing prices are sensitive to financing costs through the price-to-rent condition (schematically):

$$\frac{R_t}{P_t} \approx (\text{after-tax mortgage rate}) - \text{expected appreciation} + \text{tax/maintenance terms}.$$

Mortgage-market segmentation implies that, for homes just above P_t^* , the relevant financing rate (or weighted-average cost of funds) is higher. If supply is not perfectly elastic in the short run and the housing services flow is similar, the reduction in financing cost when a house becomes eligible for conforming 80% LTV credit should be capitalized into a higher transaction price.

Intuition 2: the CLL change is a micro-level credit-supply shift that is plausibly exogenous to local conditions

Annual CLL changes mechanically shift eligibility cutoffs, generating variation in credit access/cost that is not chosen to target specific neighborhoods in the data. This is why the design can control for granular local conditions while still moving the financing constraint.

Intuition 3: why the elasticity is “local” and smaller than some aggregate estimates

The paper estimates capitalization *only* for transactions in a tight neighborhood around the CLL-related cutoff and for borrowers whose relevant margin is moving between conforming and jumbo (or conforming plus higher-cost down payment financing). Thus, the resulting interest-rate-to-price elasticity is a local partial-equilibrium object and need not match elasticities derived from aggregate variation that bundles multiple channels and general-equilibrium feedback.